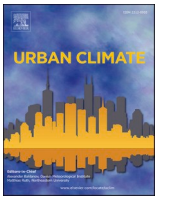
Urban Climate 55 (2024) 101967



Contents lists available at ScienceDirect

Urban Climate

journal homepage: www.elsevier.com/locate/uclim





Creating comfortable outdoor environments: Understanding the intricate relationship between sound, humidity, and thermal comfort

Zeming Qin [a,b], Bairu Lu [a,b], Wenqiang Jing [a,b,c,*], Yingdi Yin [a,b], Lefang Zhang [a,b], Xinqi Wang [a], Wei Luo [a], Jin Zhang [a], Jing Qiu [a], Yuting Dong [a,b]

[a] Xi 'an Eurasia University, School of Human Settlements and Civil Engineering, Xi 'an, Shaanxi 710065, China
[b] Engineering Research Center of Urban Intelligent Construction, Universities of Shaanxi Province, China
[c] School of Human Settlements and Civil Engineering, Xi'an Jiaotong University, Xi'an 710049, China

ARTICLE INFO

Keywords:
Outdoor thermal comfort
Cold regions
Sound levels
Humidity effects
Subjective assessments

ABSTRACT

This study provides insight into the impact of interactive sound and humidity environments on outdoor thermal comfort in open spaces in cold regions. Executed in Xi'an, China, the research involved a synthesis of questionnaire-based subjective assessments and meteorological data collection. Four pivotal findings emerged from the study. Initially, environments characterized by concurrent acoustic and humid conditions demonstrated a proportional increase in Thermal Sensation Vote (TSV) corresponding to rising humidity levels. Secondly, in these environments, an increase in sound levels was linked to an escalated thermal sensitivity among individuals, with the most significant sensitivity, quantified at 0.1769, observed at sound intensities between 60 dB (A) and 80 dB (A). Thirdly, the influence of ambient sound on TSV and Thermal Comfort Vote (TCV) was observed to vary with ambient humidity. Variations in sound levels had a pronounced effect on both TSV and TCV within a humidity range of 25% RH to 70% RH, but this correlation dissipated in higher humidity conditions (70% RH to 85% RH). Finally, the study indicates that environments with lower sound levels contribute to the mitigation of extreme thermal stress, particularly when ambient humidity is maintained between 25% RH and 70% RH.

1. Introduction

The onset of urbanization has introduced numerous environmental dilemmas, including the urban heat island effect (Heaviside et al., 2017)and the widespread occurrence of hot and humid conditions in cities (Zhang et al., 2023a). These challenges significantly deter urban residents from engaging in outdoor activities, consequently impacting their physical and mental health (Potchter et al., 2022). Establishing an outdoor environment that is both thermally comfortable and inviting is pivotal in motivating residents to actively engage in outdoor activities (Kumar and Sharma, 2020; Zhang et al., 2023b). Thermal comfort is a complex concept, influenced by several environmental factors like the acoustic environment (Favero et al., 2023) and humidity (Al-Atrash and Al-Ayyoub,

* Corresponding author at: Xi 'an Eurasia University, School of Human Settlements and Civil Engineering, Xi 'an, Shaanxi 710065, China.
E-mail addresses: qinzeming@eurasia.edu (Z. Qin), lubairu@eurasia.edu (B. Lu), jing_wenqiang@163.com (W. Jing), yinyingdi@eurasia.edu (Y. Yin), zhanglefang@eurasia.edu (L. Zhang), wangxinqi@eurasia.edu (X. Wang), luowei@eurasia.edu (W. Luo), zhangjin@eurasia.edu (J. Zhang), qiujing@eurasia.edu (J. Qiu), dongyuting@eurasia.edu (Y. Dong).

https://doi.org/10.1016/j.uclim.2024.101967
Received 28 July 2023; Received in revised form 22 April 2024; Accepted 3 May 2024
Available online 14 May 2024
2212-0955/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

2023). Analyzing these multifaceted environmental factors and their impact on urban residents' outdoor thermal comfort is crucial for improving their overall health and well-being.

In 1972, Hardy AC identified humidity as a crucial factor affecting thermal comfort (AC, 1972). This insight has led to further research into the impact of humidity on thermal comfort (Acero et al., 2022; Amaripadath et al., 2023; Yao et al., 2018; Yuan et al., 2023). Various studies have focused on humidity's effect on outdoor thermal comfort and sensation (AC, 1972). Yu et al. found that high humidity levels impair evaporative cooling, leading to increased thermal stress, reduced thermal comfort, and heightened thermal sensation (Yuan et al., 2023). Amaripadath et al. noted significant human responses to high humidity, especially when relative humidity exceeded 70% and temperatures increased (Amaripadath et al., 2023). In Juan A. Acero's study, maintaining a humidity range of 70% RH to 90% RH kept the mean Physiological Equivalent Temperature (PET) level within Singapore's acceptable thermal comfort range throughout the day and night (Acero et al., 2022). Jiawei Yao's research in winter Shanghai showed that a humidity range of 70% RH to 90% RH led to most respondents being comfortable with the thermal environment, with PET temperatures between 7.7 °C and 21.8 °C (Yao et al., 2018).

Contradicting earlier assertions that outdoor thermal comfort is unaffected by sound (Fanger et al., 2007), recent inquiries have begun to unearth the interconnection between sound and thermal comfort (Du et al., 2023a; Jin et al., 2020; Lau and Choi, 2021; Nicolas Pellerin, 2003; Yang and Moon, 2018). Du et al. demonstrated that escalated sound levels intensify stress and heat sensation, thereby undermining thermal comfort (Du et al., 2023a). Kevin and Choi's analysis of perceived acoustics and outdoor thermal comfort during hot summers indicated that tranquil and visually pleasing outdoor settings enhance thermal tolerance and minimize thermal sensitivity (Lau and Choi, 2021). Jin et al.'s findings suggest that although sound does not influence thermal sensation in cooler conditions, it contributes to thermal discomfort (Jin et al., 2020). Pellerin and Candas' research into the combined effects of sound and temperature on discomfort revealed that sound modifies thermal comfort in warmer climates (Nicolas Pellerin, 2003). Yang et al.'s study on the effects of sound and thermal conditions on subjective environmental perception concluded that while sound does not impact thermal sensation, it negatively influences thermal comfort as sound intensity increases (Yang and Moon, 2018).

Prior research has predominantly focused on the effects of individual environmental factors on thermal comfort, such as the singular impacts of acoustic and humid conditions. In contrast, residents typically face a spectrum of environmental interactions (Lai et al., 2020). Research by Bin et al. indicates that the aggregate physical and psychological stress encountered in a comprehensive environmental context notably exceeds the influence exerted by any single environmental element (Yang et al., 2022). Furthermore, the assessment of outdoor thermal comfort is governed by a blend of psychological and physiological factors (Lam et al., 2021). This implies that an analysis confined to thermal comfort within a singular environmental aspect inadequately reflects the authentic adaptive thermal responses of individuals in the intricate milieu of urban settings.

This study, through a questionnaire survey and the collection of meteorological data at a university in Xi'an, aimed to investigate the changes in outdoor thermal comfort experienced by individuals in an environment influenced by the interplay of sound and humidity. The research objectives were specifically to: (1) evaluate the influence of acoustic environment variations on outdoor

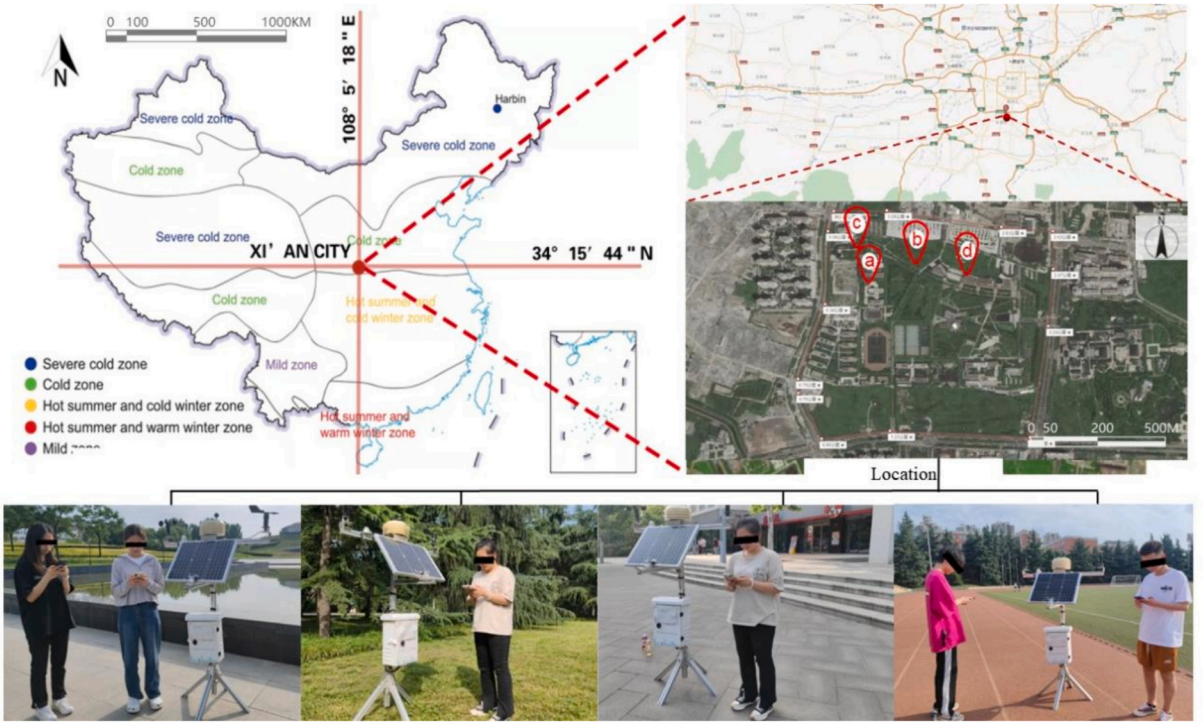


Fig. 1. Location map of the measurement location.

thermal comfort, and (2) analyze the effects of humidity level changes on outdoor thermal comfort in a consistent sound environment. The findings from this research not only deepen the understanding of the interrelated effects of sound and humidity on outdoor thermal comfort and sensation but also provide essential insights for architects, urban planners, and policymakers engaged in the design and strategic planning of outdoor environments (Guan et al., 2020; Ma et al., 2021).

2. Experimental

2.1. Experiment site

Xi'an, positioned within the warm temperate semi-humid monsoon climate zone, is characterized by moderate rainfall and four distinct seasons. Analysis of climate data spanning the past decade indicates an average annual temperature of 13.3 °C, annual precipitation fluctuating between 507.7 mm and 719.8 mm, and an average humidity level of 69.6% RH. The research was undertaken at a university located in Xi'an (108°922′ 765″E, 34.177′192″ N). The university's campus, sprawling over approximately 80 ha, is replete with verdant vegetation and a richly diverse outdoor cultural landscape. This setting presents an ideal backdrop for investigating the dynamics of sound, humidity, and thermal comfort in campus environments. The study was conducted across four representative campus locations: an open square, a playground, an area adjacent to a pool, and a lawn section. These sites typify the standard spatial layouts found on the campus (Fig. 1).

2.2. Experiment design

The study will conduct experiments from 11:30 to 14:30 every day from June 1 to September 1, 2021. In case of bad weather, the experiment will be postponed. A total of 20 volunteers, comprising 10 men and 10 women, participated in the experiment. To minimize the effect of gender differences on the results, the volunteers were organized into two groups, each containing 5 men and 5 women. Before the experiment, the volunteers were briefed on the study's purpose, methods, and procedures to ensure they fully understood and agreed to participate. The two groups of volunteers alternately carried out the experiments, visiting the four selected sites in a predetermined order. At each site, they rested for 5–10 min in a standing position, observed the environment, and completed questionnaires.

Parallel to these activities, environmental parameters such as air temperature, relative humidity, and sound decibel levels were measured at a standardized height of 1.2 m using a mobile weather station (Zhang et al., 2023a). The specifics regarding the measurement instrument, including its model, operational range, accuracy, and appropriate environmental conditions, are enumerated in Table 1. To ensure accuracy and mitigate the interference of ancillary sound sources in the sound decibel readings, a minimum of 20 measurements were recorded at each site, with the conclusive data being the mean of these readings (Yin et al., 2022).

2.3. Questionnaires

The questionnaire for the study consisted of three parts. The first section collected the participants' personal information, including gender, age, health status, and clothing type. Participants unwilling to disclose their personal details due to privacy concerns were excluded from the test. The second section required participants to evaluate the Thermal Sensation Vote (TSV) and Thermal Comfort Vote (TCV) of the current environment, as well as their perceptions of the acoustic and humidity environments, using the Likert 7-point scale to reflect their satisfaction and subjective comfort levels. The final section involved the investigators documenting the environmental physical parameters during the test, such as wind speed, sound, black globe temperature, humidity, and illumination.

In the course of the experiment, a total of 1700 questionnaires were completed, out of which 1603 were adjudged valid for inclusion in the study. Given the university campus setting of the test site, the participant demographic was exclusively college students,

Table 1
Basic parameters of the measuring instruments used.

Instrument	Measuring range	Measurement accuracy	Equipment size	Work environment
Noise detector (noise0501)	30db-120db	±0.5db	8*5*12 cm	Air temperature: −40% ~ +60% Relative humidity: 25% ~ 90%
Wind speed detector (wind0501)	0.2 m/s − 10 m/s	±0.02 m/s	8*5*12 cm	-10 °C ~ +50 °C
Black ball temperature / Humidity detector	Illumination: 0–65,535 lx Humidity: −40 °C ~ +125 °C Black ball: −10 °C ~ +85 °C	Humidity: ±0.3 °C , ± 2%RH Black ball: ±0.5 °C	12*8*6 cm	

predominantly aged 19 to 23 years.

2.4. Procedures

With the continuous development of outdoor thermal comfort research, scholars often use PET, Standard Effective Temperature (SET), Perceived Equivalent Temperature (TEP) and Uuniversal Thermal Climate Index (UTCI) to construct outdoor thermal comfort models (Davtalab et al., 2020). Notably, PET has been increasingly adopted in recent Outdoor Thermal Comfort (OTC) assessments due to its capacity for enabling comparative evaluations of thermal comfort across diverse environmental settings (Yin et al., 2022). In the scope of this study, PET was specifically selected to forecast thermal comfort levels. To calculate the PET values for all participants, the Rayman model was methodically employed throughout the research timeframe (Acero et al., 2022).

3. Results

3.1. Descriptive analysis

To study the effects of off-chamber humidity and sound on the human thermal sensation and comfort sensation, grade them. According to the international standard ISO 7730 and ISO 1996-2:2007, the relative humidity and sound of the environment will be comfortable within a certain range, beyond which the range will have a negative impact.

In the detailed analysis of both the questionnaire data and the actual meteorological observations, it became evident that the subjects' assessment of humidity primarily ranged across four levels: somewhat dry, moderate, wet, and very wet. Concerning sound levels, the responses were mainly categorized as quiet, moderate, and noisy. Utilizing the environmental data collected and the participants' input, humidity was methodically divided into four grades: 25% RH to 40% RH, 40% RH to 55% RH, 55% RH to 70% RH, and 70% RH to 85% RH. Likewise, the sound levels were systematically categorized into three distinct ranges: 20 to 40 dB (A), 40 to 60 dB (A), and 60 to 80 dB (A).

Table 2 shows that the significance of the indicators under TSV, TCV, PET, Relative humidity and Equivalent continuous sound level. The data analysis conclusively demonstrated significant interrelations among the TSV, TCV, PET, humidity intervals, and sound decibel intervals. This denotes that fluctuations in humidity intervals notably influence the TSV, TCV, PET, and sound decibel intervals, while changes in sound decibel intervals reciprocally affect the TSV, TCV, PET, and humidity intervals. Accordingly, this paper embarks on a comprehensive scientific inquiry into the mutual interactions and variations of humidity and sound decibel intervals, assessing their collective impact on thermal comfort and sensation metrics.

3.2. Comprehensive influence of sound environment on thermal comfort under the same humidity range

3.2.1. The humidity interval is within 25% RH ~ 40% RH, and the effect of acoustic environment on MTSV / MTCV

Fig. 2 (A) depicts the subjective MTCV in an acoustic thermal setting, with humidity levels ranging from 25% RH to 40% RH. The survey's findings showed that the MPET extended from 19 °C to 48 °C, while the MTCV was measured between −3 and 0.5, averaging at −1.469. This average value indicates a consensus among respondents that the ambient thermal environment was generally perceived as uncomfortable. Notably, in the MPET range of 38 °C to 40 °C, there was a noticeable downward trend in MTCV as MPET increased, suggesting that rising temperatures led to heightened perceptions of discomfort and diminished thermal comfort.

The analysis of Fig. 2(A) uncovers a positive correlation between MTCV and sound decibel interval at selective MPET values. Specifically, at MPET values such as 30 °C, 32 °C, and 37 °C, a rise in the sound decibel interval is associated with an increased MTCV. Conversely, at MPET values like 38 °C and 40 °C, a negative correlation is evident, where higher sound decibel intervals correspond to lower MTCV values. Notably, when MPET surpasses 42 °C, the sound decibel interval no longer significantly affects MTCV, especially

Table 2
The significance of the indicators under TSV, TCV, PET, Relative humidity and Equivalent continuous sound level.

		TSV	TCV	PET	Humidity interval	Sound decibel interval
TSV	correlation	1	−0.646**	0.806**	0.052*	−0.083**
	Sig.(Double tail)		0	0	0.038	0.001
	N	1603	1603	1603	1603	1603
TCV	correlation	−0.646**	1	−0.741**	0.196**	0.051*
	Sig.(Double tail)	0		0	0	0.043
	N	1603	1603	1603	1603	1603
PET	correlation	0.806**	−0.741**	1	−0.311**	0.043*
	Sig.(Double tail)	0	0		0	0.087
	N	1603	1603	1603	1603	1603
Humidity interval	correlation	0.052*	0.196**	−0.311**	1	−0.233**
	Sig.(Double tail)	0.038	0	0		0
	N	1603	1603	1603	1603	1603
Sound decibel interval	correlation	−0.083**	0.051*	0.043*	−0.233**	1
	Sig.(Double tail)	0.001	0.043	0.087	0	
	N	1603	1603	1603	1603	1603

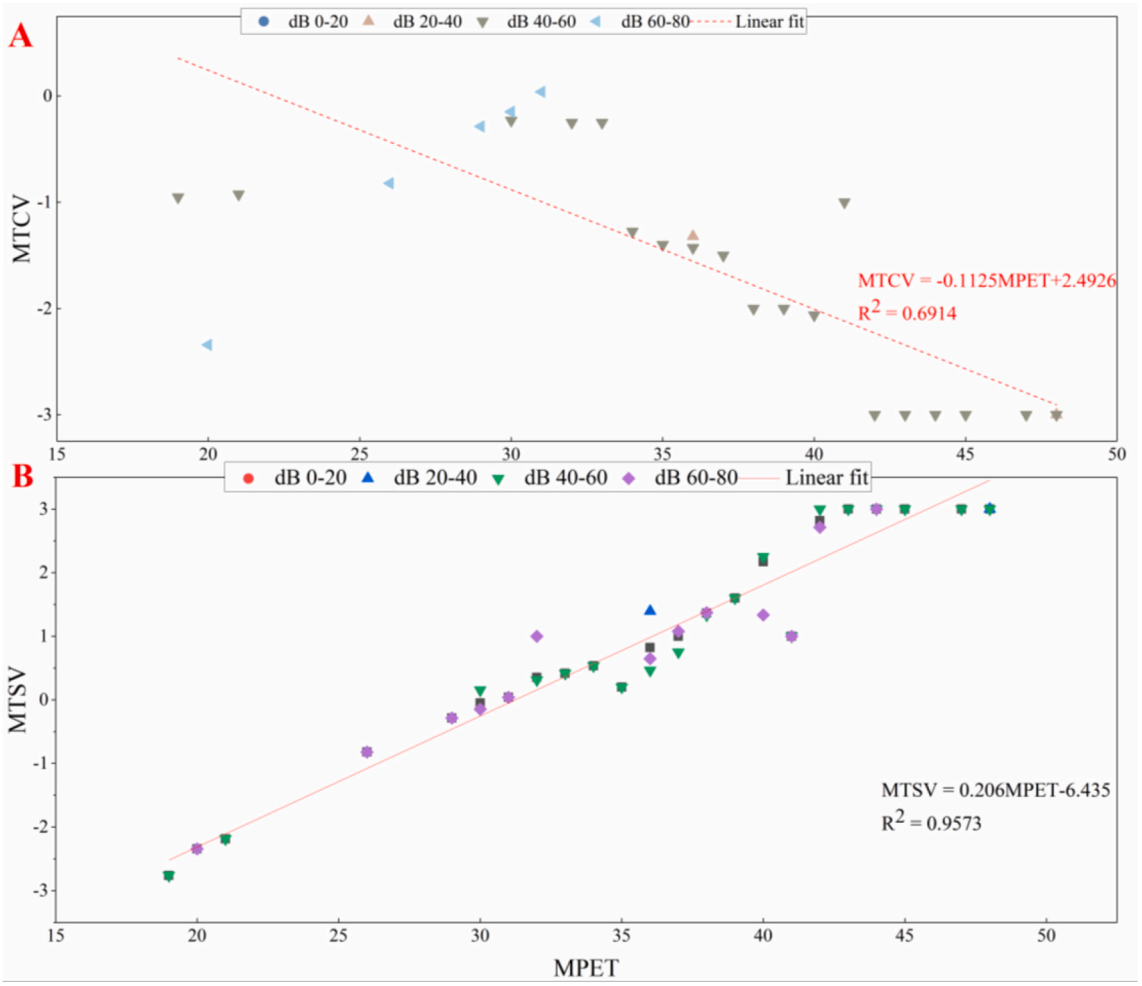


Fig. 2. The range of humidity is within 25% RH ~ 40% RH, and the influence of acoustic environment on MTSV and MTCV.

at an MTCV value of −3. These observations illustrate that the relationship between sound decibel interval and MTCV is intricately linked to both the specific humidity level and MPET.

Fig. 2 (B) showcases the subjective TSV within the acoustic thermal environment under humidity conditions ranging from 25% RH to 40% RH. The analysis of the survey data revealed that the MPET extended across a spectrum from 19 °C to 48 °C. In this context, the MTSV was observed to oscillate between −2.7 and 3, averaging at 0.820. This average indicates that the respondents generally characterized the surrounding thermal environment as slightly warm.

The analysis of Fig. 2(B) demonstrates a significant correlation between sound decibel intervals and the MTSV at certain MPET thresholds. Specifically, at MPET levels of 32 °C and 37 °C, an elevation in sound decibel intervals is linked to an increase in MTSV values. Conversely, at MPET values such as 40 °C and 42 °C, there exists a negative correlation, with higher sound decibel intervals leading to lower MTSV values. Notably, beyond MPET values of 43 °C, the influence of sound decibel intervals on MTSV diminishes, particularly when the MTSV reaches a value of 3.

3.2.2. The humidity interval is within 40% RH ~ 55% RH, and the effect of acoustic environment on MTSV / MTCV

Fig. 3 (A) portrays the subjective MTCV within an acoustic thermal environment in the humidity range of 40% RH to 55% RH. The survey data reveal that the MPET extended from 20 °C to 46 °C, with MTCV values ranging from −3 to 0.6, and an average of −1.026, suggesting a general perception of slight discomfort in the thermal environment. At MPET values of 32 °C and 38 °C, the thermal comfort was found to be higher in the lower sound decibel interval (40–60 dB (A)) than in the higher interval (60–80 dB (A)) with the increase in MPET. Beyond an MPET of 43 °C, the influence of the sound decibel interval on MTCV was not significant, particularly when MTCV reached −3. Overall, it was observed that with rising MPET, MTCV evaluation decreased, indicating that lower sound decibel intervals were associated with higher thermal comfort than higher intervals at elevated temperatures.

Fig. 3 (B) showcases the subjective MTSV within an acoustic thermal setting, characterized by humidity levels ranging from 40% RH to 55% RH. The survey data revealed that the MPET ranged from 20 °C to 46 °C, with MTSV values oscillating between −2.2 and 3,

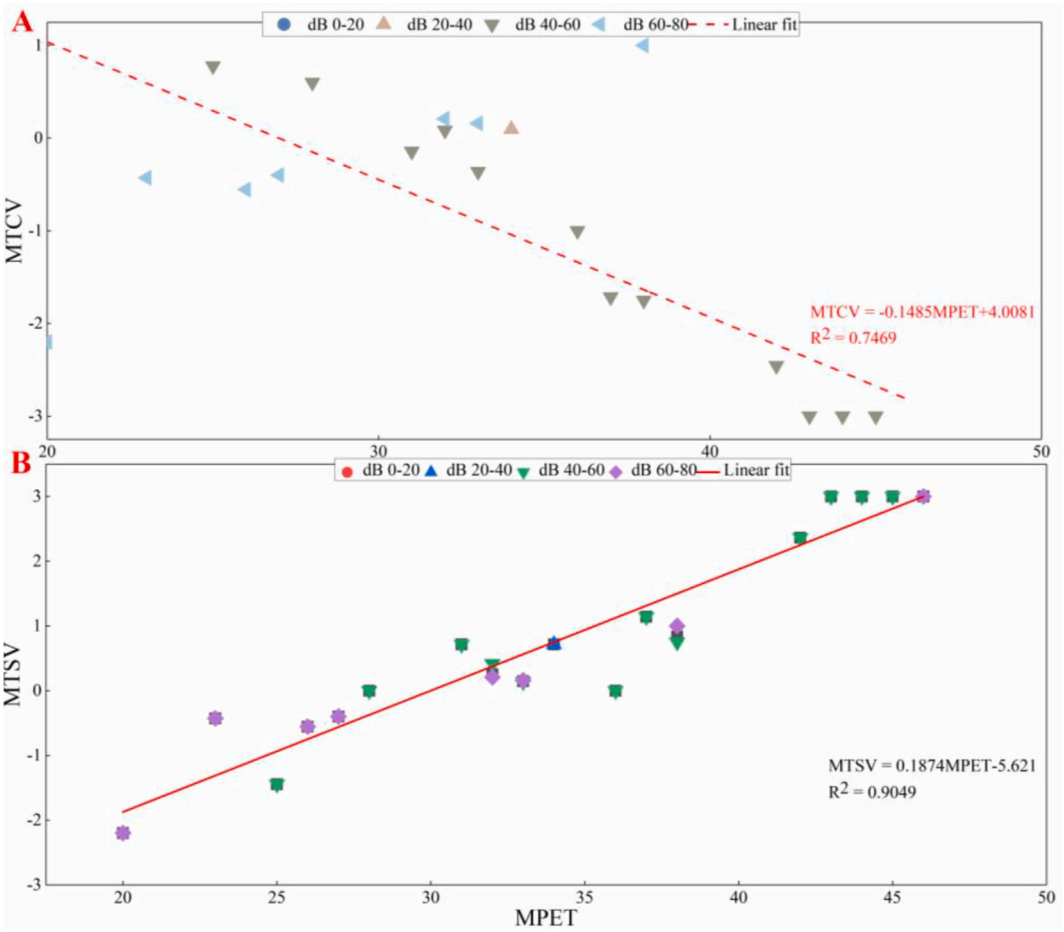


Fig. 3. The range of humidity is within 40% RH ~ 55% RH, and the influence of acoustic environment on MTSV and MTCV.

averaging at 0.730. This average suggests that respondents predominantly perceived the thermal environment as slightly warm. Notably, in the MPET range of 33 °C to 34 °C, a notable increase in MTSV was observed alongside the rising MPET, indicating that as temperatures escalate, the perception of warmth and the intensity of the thermal environment also increases.

The analysis of Fig. 3(B) elucidates a positive correlation between the sound decibel interval and the MTSV at select MPET values. Notably, this positive correlation is observed at MPET values of 33 °C and 38 °C, where a higher sound decibel interval aligns with an increased MTSV. In contrast, at an MPET value of 32 °C, a negative correlation is evident, with higher sound decibel intervals leading to reduced MTSV values. Furthermore, beyond MPET values of 43 °C, the sound decibel interval ceases to have a significant effect on MTSV, particularly when MTSV reaches the level of 3.

3.2.3. The humidity interval is within 55% RH ~ 70% RH, and the effect of acoustic environment on MTSV / MTCV

Fig. 4 (A) portrays the subjective MTCV within an acoustic thermal environment, characterized by a humidity range of 55% RH to 70% RH. The survey results indicated that the MPET extended from 19 °C to 53 °C, with MTCV scores ranging from −3 to 1 and an average of −1.150, denoting a perception of slight discomfort in the thermal environment. At MPET values of 25 °C, 38 °C, and 39 °C, there was a notable trend where the MTCV was higher in the 40–60 dB (A) sound decibel interval compared to the 60–80 dB (A) range, suggesting that lower decibel levels were associated with greater thermal comfort as temperatures increased. When the MPET exceeded 45 °C, the influence of sound decibel intervals on MTCV became insignificant, particularly at an MTCV score of −3. Overall, the analysis revealed that with the progression of MPET, the evaluation of MTCV gradually declined, implying that thermal comfort was higher at lower sound decibel intervals in warmer temperatures.

Generally, higher sound levels within the 60 to 80 dB (A) range are associated with reduced thermal comfort, as indicated by larger negative values in MTCV at these sound intervals. While the influence of different sound levels on MTCV is relatively minimal at lower MPET values, this impact becomes more pronounced with increasing MPET. This trend underscores that the effect of sound on individuals' comfort is more significant at elevated temperatures, highlighting the increased sensitivity to auditory stimuli in warmer environments.

Fig. 4 (B) provides a visual representation of the subjective MTSV in an acoustic thermal environment, specifically within the 55%

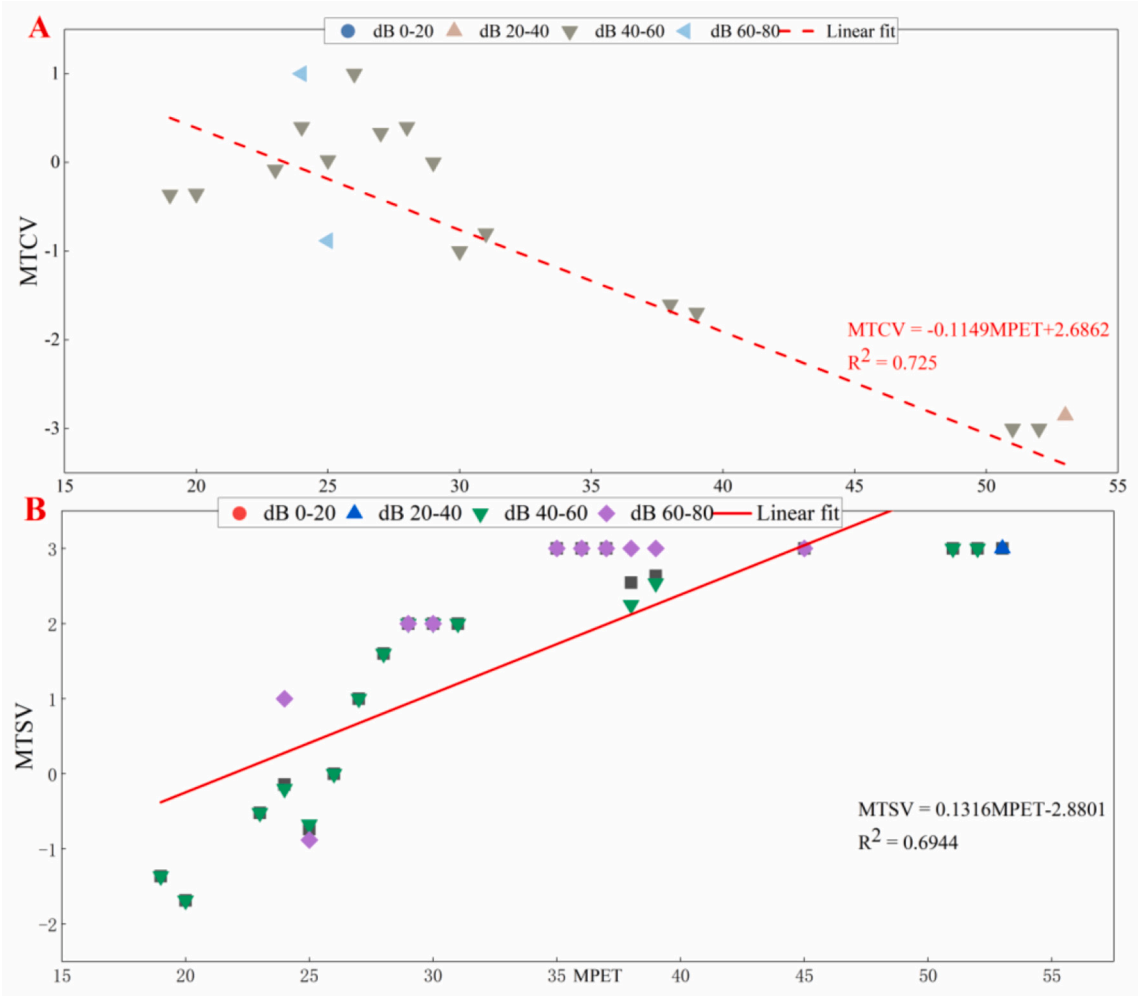


Fig. 4. The range of humidity is within 55% RH ~ 70% RH, and the influence of acoustic environment on MTSV and MTCV.

RH to 70% RH humidity range. Survey data indicated that the MPET fluctuated from 19 °C to 53 °C, with TSV values spanning −1.6 to 3 and averaging at 1.517, reflecting a perception of warmth in the surrounding thermal environment. Particularly at MPET values of 24 °C, 38 °C, and 39 °C, there was an observed trend of increasing MTSV with the rise in MPET, suggesting that higher sound decibel levels are associated with a warmer thermal environment perception. When the MPET exceeded 45 °C, the influence of sound decibel intervals on MTSV was not substantial, especially once the MTSV attained a value of 3.

3.2.4. The humidity interval is within 70% RH ~ 85% RH, and the effect of acoustic environment on MTSV / MTCV

Fig. 5 (A) shows the subjective MTCV within an acoustic thermal environment in the humidity range of 70% RH to 85% RH. Survey data indicate that the MPET was within the range of 19 °C to 53 °C, with MTCV spanning from −3 to 1.2 and averaging −1.042, indicating a general perception of slight discomfort in the thermal environment. At MPET values of 17 °C, 19 °C, 24 °C, 29 °C, 31 °C, and 38 °C, there existed a positive correlation between the sound decibel interval and MTCV. Conversely, at MPET levels of 25 °C, 28 °C, 32 °C, 35 °C, and 36 °C, a negative correlation between the sound decibel interval and MTCV was noted. When MPET exceeded 45 °C, the influence of the sound decibel interval on MTCV was negligible, particularly when MTCV reached a value of −3.

The analysis of the Fig. 5(A) underscores the significant influence of humidity on thermal comfort. In a given humidity context, when the sound environment shifts from a range of 20–40 dB (A) to 40–60 dB (A) or even 60–80 dB (A), there is a noticeable effect on the MTSV among individuals. The link between MTSV and the sound environment, while not markedly significant in humidity levels of 25% RH to 40% RH, becomes increasingly evident in higher humidity settings of 40% RH to 55% RH, 55% RH to 70% RH, and 70% RH to 85% RH. In these higher humidity conditions, the population exhibits greater sensitivity to variations in the sound environment, highlighting the interplay between auditory stimuli and thermal comfort perceptions.

Fig. 5 (B) showcases the subjective MTSV in an acoustic thermal setting, within the humidity range of 70% RH to 85% RH. The survey data delineated that the MPET covered a spectrum from 17 °C to 53 °C, with MTSV values spanning −2 to 3 and an average of 1.301, indicating a general perception of the thermal environment as slightly warm. Specifically, at MPET values of 25 °C, 29 °C, and

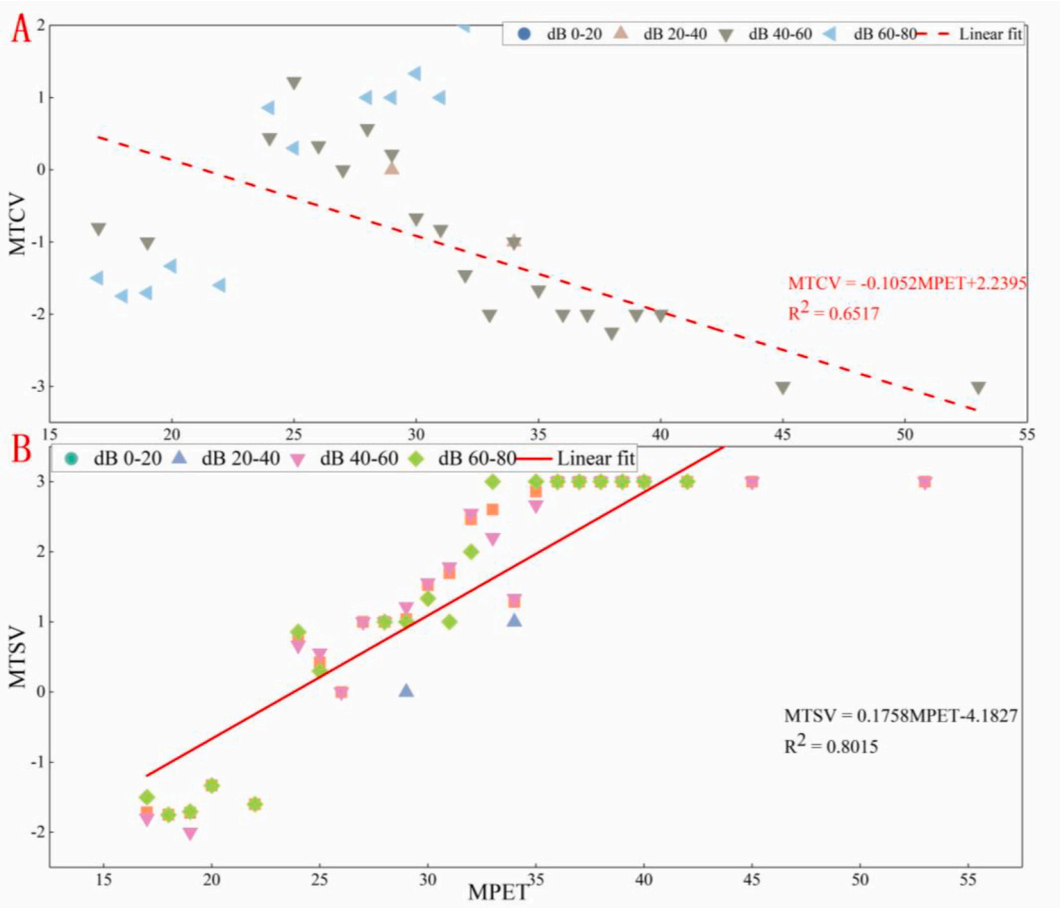


Fig. 5. The range of humidity is within 70% RH ~ 85% RH, and the influence of acoustic environment on MTSV and MTCV.

32 °C, the MTSV was notably higher in the 40–60 dB (A) sound interval than in the 60–80 dB (A) interval, denoting that lower sound levels were linked to an increased sensation of warmth as MPET rose. In contrast, at MPET levels of 17 °C, 19 °C, 24 °C, 33 °C, and 35 °C, the perception of warmth was more intense in the 60–80 dB (A) interval. When MPET exceeded 36 °C, the sound decibel interval had negligible impact on MTSV, especially when MTSV reached 3.

The interplay of sound environments and human thermal sensation is considerably significant. An increase in sound decibels correlates with changes in the body's thermal sensation. Specifically, heightened sound levels can lead to reduced thermal sensation, as exemplified by the MTSV registering at 2.2 in 20% RH humidity with 20–40 dB (A), 1.2 at 40–60 dB (A), and 0.7 at 60–80 dB (A). Furthermore, different humidity settings also markedly influence MTSV. In environments with humidity levels of 40%RH to 55%RH and 55%RH to 70%RH, MTSV tends to be lower compared to environments of 70%RH to 85%RH, under similar MPET conditions. Within consistent sound and humidity conditions, MTSV generally increases with rising MPET. However, there are notable exceptions, such as in a 25% RH to 40% RH humidity environment, where an increment in MPET from 24 °C to 25 °C resulted in a decrease in MTSV.

3.2.5. The variation pattern of MTSV/MTCV under comprehensive conditions

Based on the analysis of Figs. 2(A), 3(A), 4(A) and 5(A), it can be seen that in all humidity ranges, when MPET exceeds 45 °C, the sound decibel range has no impact on MTCV, that is, the MTCV value remains unchanged at −3. In the humidity ranges extending from 25% RH to 70% RH, fluctuations in the sound decibel interval correlate with variations in MTCV. Conversely, within the higher humidity domain of 70% RH to 85% RH, the correlation between the sound decibel range and MTCV becomes less pronounced, indicating a diminished sensitivity to sound decibel changes in terms of thermal comfort under these conditions.

A comprehensive analysis of Fig. 2(B), 3(B), 4(B) and 5(B) shows that when the humidity range is between 25%RH and 70%RH, MPET exceeds 43 °C, and the sound decibel range has no effect on MTSV, that is, the MTSV value remains unchanged at 3. Conversely, in a humidity interval of 70% RH to 85% RH, the sound decibel interval shows no significant effect on MTSV when MPET surpasses 36 °C. In lower humidity conditions of 25% RH to 40% RH, and at MPET values under 37 °C, a positive correlation exists between the sound decibel interval and MTSV. As MPET increases, this correlation turns negative. Within the 40% RH to 55% RH and 55% RH to 70% RH humidity ranges, at lower MPET levels (below 32 °C), a negative correlation is noted between the sound decibel interval and

MTSV, while at higher MPET values, the correlation becomes positive. This trend is in stark contrast to the pattern observed in the 25% RH to 40% RH range. However, in the higher humidity range of 70% RH to 85% RH, the relationship between the sound decibel interval and MTSV does not follow this distinct pattern.

3.3. The comprehensive influence of humidity on thermal comfort under the same sound decibel interval

3.3.1. The influence of humidity on MTSV / MTCV in the acoustic environment within 20–40 dB (A)

Fig. 6 (A) elucidates the subjective MTCV in a humid environment at sound levels between 20 and 40 dB (A). It is observed that MTCV consistently decreases with the escalation of MPET, exhibiting significant fluctuations, which highlights the considerable influence of humidity on thermal comfort. The survey data delineates that MTCV values predominantly fell below zero, ranging from 0.1 to −3, with an average of −1.43, indicating that the environment was perceived as uncomfortable by respondents in these humidity and decibel conditions.

The data analysis across various humidity ranges reveals that an increase in MPET consistently results in a decrease in MTCV, indicating that higher MPET levels are linked to reduced thermal comfort in outdoor settings. At an MPET of 34 °C, there is a distinct negative correlation between humidity intervals and MTCV, with higher humidity leading to lower MTCV. When MPET surpasses 48 °C, any changes in humidity intervals cease to have an impact on MTCV, which consistently remains at −3, reflecting a stable perception of discomfort irrespective of humidity variations at these higher temperatures.

Fig. 6 (B) demonstrates the subjective MTSV within a humid environment at sound levels between 20 and 40 dB (A). The data reveals a trend of increasing MTSV in correspondence with rising MPET. The MTSV values, predominantly above zero, ranged from 0 to 3, averaging at 1.62, suggesting that respondents perceived the environment as warm. At an MPET of 34 °C, there is a clear positive correlation between humidity intervals and MTSV, with MTSV reaching its peak in the 55% RH to 70% RH humidity range before diminishing as humidity increases further to 70% RH to 85% RH. When MPET exceeds 48 °C, alterations in humidity intervals do not substantially affect MTSV, maintaining it consistently at a value of 3.

3.3.2. The influence of humidity on MTSV / MTCV in the acoustic environment within 40–60 dB (A)

Fig. 7 (A) delineates the MTCV in a humid environment at sound levels between 40 and 60 dB (A). The correlation coefficient between MTCV and MPET is established at 0.763, affirming a substantial correlation between these two factors. Survey data reveals that MTCV evaluations exhibited a broad range, extending from −3 to 1.2, with an average value of −1.311. This data reflects a general

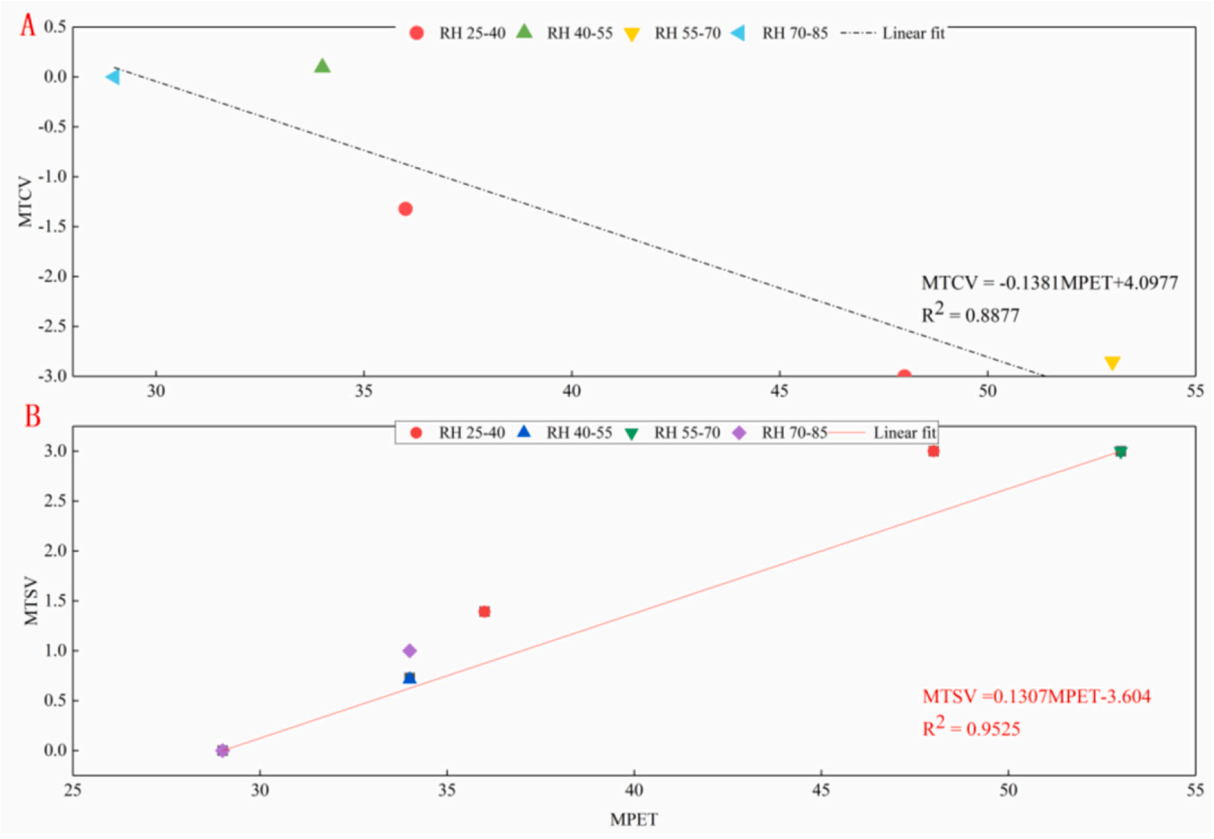


Fig. 6. The decibel range of 20–40 dB (A), the influence of humid environment on MTSV and MTCV.

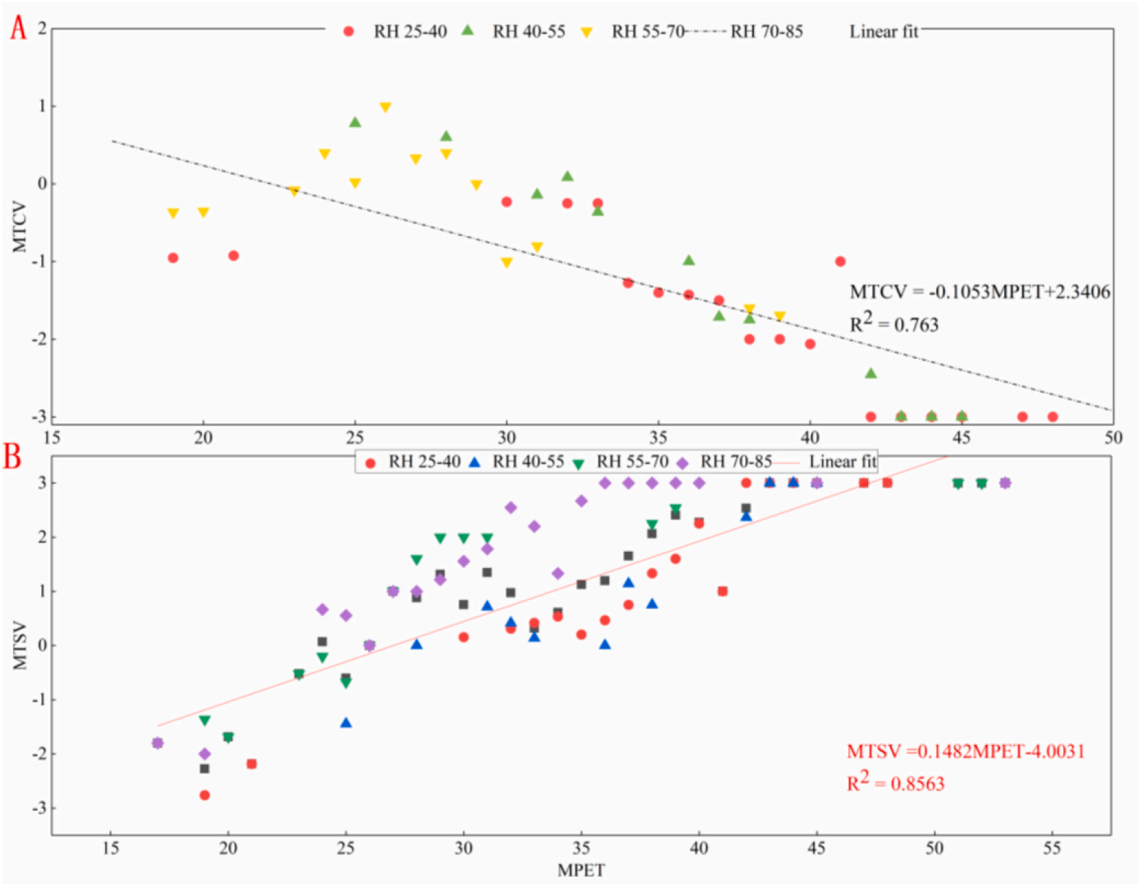


Fig. 7. The decibel range of 40–60 dB (A), the influence of humid environment on MTSV and MTCV.

perception among respondents of the thermal environment being moderately uncomfortable.

Analysis of Fig. 7(A) reveals that the correlation between changes in the humidity interval and MTCV is not uniformly clear. However, within the humidity range of 25% RH to 70% RH, there is a noticeable relationship between these factors. Specifically, at lower MPET values, below 37 °C, a negative correlation exists between the change in humidity interval and MTCV. Conversely, when MPET exceeds 38 °C, there is a positive correlation observed between humidity interval changes and MTCV. Beyond an MPET of 43 °C, the impact of humidity interval changes on MTCV diminishes, with MTCV consistently remaining at −3, indicating a plateau in thermal comfort perception at these higher temperatures and humidity levels.

Fig. 7 (B) provides insight into the subjective MTSV in a humid environment at sound levels of 40 to 60 dB (A). The correlation coefficient of 0.8563 between MPET and MTSV underscores a significant correlation. Survey data indicate that MTSV ranged extensively, from −2.7 to 3, with an average value of 1.140, suggesting that the respondents perceived the environment as relatively warm. In scenarios of changing MPET, particularly at values of 29 °C and 42 °C, there is a discernible negative correlation between humidity interval changes and MTSV, with lower humidity levels resulting in higher thermal sensation. Conversely, at MPET values of 25 °C, 32 °C, 34 °C, 35 °C, 37 °C, 39 °C, and 40 °C, an increase in humidity intervals is associated with a rise in MTSV. When MPET surpasses 43 °C, humidity interval changes cease to have a marked impact on MTSV, maintaining it consistently at 3.

3.3.3. The influence of humidity on MTSV / MTCV in the acoustic environment within 60–80 dB (A)

Fig. 8 (A) depicts the MTCV in a humid environment with sound levels in the range of 60 to 80 dB (A). The data presents a correlation coefficient of 0.6823, signifying a moderate yet significant correlation between environmental factors and MTCV. Survey findings show that MPET varied from 17 °C to 46 °C, with MTCV evaluations fluctuating between −3 and 1, averaging −1.011, indicating a general perception of slight discomfort. The analysis indicates that at MPET values such as 24 °C, 26 °C, 31 °C, 33 °C, 37 °C, 38 °C, and 40 °C, higher humidity intervals correlate with lower MTCV values. Conversely, at MPET values of 25 °C and 35 °C, an increase in humidity interval positively affects MTCV. When MPET exceeds 41 °C, changes in humidity intervals cease to significantly impact MTCV, which remains stable at −3.

Fig. 8 (B) illustrates the MTSV in a humid and heated environment with sound levels between 60 and 80 dB (A). The fitting analysis between MTSV and MPET results in an R2 value of 0.8356, highlighting a strong correlation between them. Survey data reveals that MPET ranged from 17 °C to 46 °C, with MTSV evaluations extending from −2.3 to 3, and an average of 0.671, indicating a perception

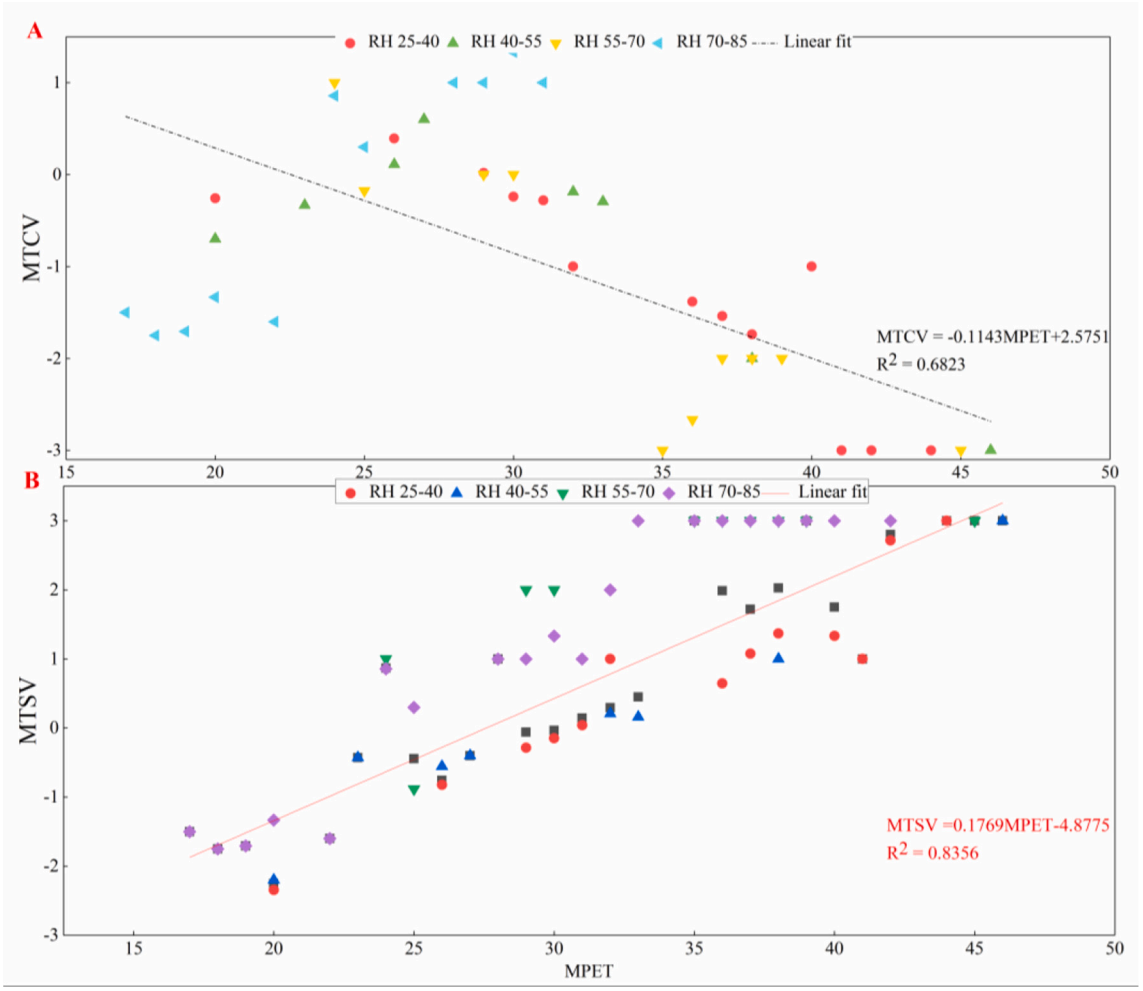


Fig. 8. The decibel range of 60–80 dB (A), the influence of humid environment on MTSV and MTCV.

of slight warmth by respondents. The figure analysis shows that at MPET values of 25 °C, 26 °C, 31 °C, 33 °C, 36 °C, 37 °C, 40 °C, and 41 °C, increases in humidity intervals are positively correlated with increases in MTSV. Conversely, when MPET exceeds 42 °C, changes in humidity intervals do not significantly impact MTSV, maintaining it at a steady value of 3, signifying consistent thermal sensation at these higher temperatures.

3.3.4. The variation pattern of MTSV/MTCV under comprehensive conditions

An integrative analysis of Figs. 6(A), 7(A), and 8(A) indicates that within decibel intervals of 40 to 80 dB (A), when the MPET exceeds 43 °C, humidity interval changes do not impact the MTCV, resulting in a stable MTCV value of −3. On the other hand, in a decibel interval of 20 to 40 dB (A), once the MPET surpasses 48 °C, the MTCV is similarly unaffected by humidity interval changes, consistently maintaining a value of −3. This pattern highlights the consistent perception of thermal discomfort in higher temperature conditions, irrespective of humidity changes, within these specific sound level ranges.

An aggregated analysis of Figs. 6(B), 7(B), and 8(B) demonstrates that within decibel intervals of 40 to 80 dB (A), when the MPET exceeds 43 °C, humidity interval changes do not significantly influence the MTSV, with MTSV remaining at a constant value of 3. Conversely, in a lower decibel range of 20 to 40 dB (A), when MPET exceeds 48 °C, humidity interval changes also do not impact MTSV, which remains stable at 3. The fitted line analysis for MTSV and MPET across these figures shows varying sensitivity levels at different decibel values. In the decibel range of 20 to 40 dB (A), the slope of the fitted line between MTSV and MPET is 0.1307, increasing to 0.1482 in the 40 to 60 dB (A) range and further to 0.1769 in the 60 to 80 dB (A) range. This trend indicates heightened sensitivity of MTSV to changes in MPET at higher decibel levels.

4. Discussion

4.1. The effect of humidity on thermal comfort at the same sound decibel interval

The current study delved into the dynamics of outdoor TSV within a population under the simultaneous influence of sound and humidity. The results from this study lend support to the theory that an increase in humidity, within an environment where sound is also a factor, elevates the outdoor TSV experienced by individuals. These findings are consistent with the empirical work conducted by Berglund (Berglund, 1998) and align with the descriptions by Chen (Chen et al., 2011).

The results of this study intriguingly revealed that the population's TSV displayed increased sensitivity in response to higher sound decibels in the outdoor environment, indicating a complex interplay between auditory and thermal stimuli. Corroborating this, Yang et al. observed that elevated environmental sound intensifies psychological stress, leading to a heightened perception of warmth (Yang et al., 2022). Additionally, Abbasi et al. highlighted that the influence of combined environmental factors is more substantial than that of individual factors alone (Abbasi et al., 2020). Consequently, with the increase in outdoor humidity, there is a notable rise in TSV. Simultaneously, the amplification of the outdoor acoustic environment further escalates this increase in TSV. Therefore, it is essential to consider both outdoor humidity and sound levels when analyzing their collective impact on thermal sensation.

In this study, we discovered that individuals in outdoor environments with lower sound levels (20 to 40 dB (A)) tended to report less extreme thermal discomfort, reflected by a TCV of −3. This observation presents a contrast to the findings of Zhu et al., who associated high humidity with increased discomfort`(Zhu et al., 2018), and Sachindra et al., who indicated a decrease in thermal comfort with rising humidity (Sachindra and Nowosad, 2022). Our findings point to the possibility that quieter environments may counterbalance the discomfort typically associated with high humidity, likely due to a decrease in psychological stress induced by lower levels of acoustic stimulation.

In the context of this study, it is critical to acknowledge the limitations posed by the sparse data samples, particularly in the 20 to 40 dB (A) range and under conditions of low decibels combined with high humidity (20 to 40 dB (A), 70% RH to 85% RH). Therefore, the conclusions drawn are more theoretical in nature and less grounded in practical application. The study brings to light the complex dynamics between auditory and thermal sensations and their potential role in improving thermal comfort and reducing discomfort. However, it is essential to recognize that these results are based on a limited scope of data analysis and carry a degree of speculation. Future research endeavors should focus on a more detailed exploration of how subtle differences in sound and humidity in outdoor settings influence human thermal comfort.

4.2. When the humidity range is the same, the sound affects the thermal comfort

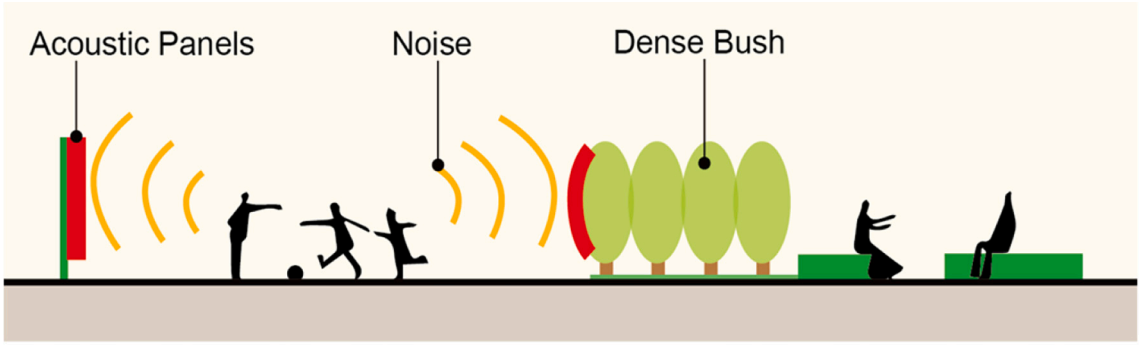
The changing relationship between outdoor sound and TSV and TCV has been a topic of complexity. Scholars have worked on this topic and obtained different results.

Several studies have elucidated the distinct relationship between sound and thermal comfort perceptions, specifically focusing on TCV versus TSV. Notably, research by Yang, Moon et al.(Yang and Moon, 2018), as well as Juang et al. and Yang et al.(Juang et al., 2010), revealed that while sound levels have a minimal impact on TSV, they exhibit a significant correlation with TCV. These studies consistently found a substantial negative correlation between increased noise levels and TCV, indicating that higher sound levels lead to a significant reduction in TCV, in contrast to their lesser influence on TSV. This underscores the nuanced impact of auditory stimuli on different aspects of thermal perception in the outdoor environment.

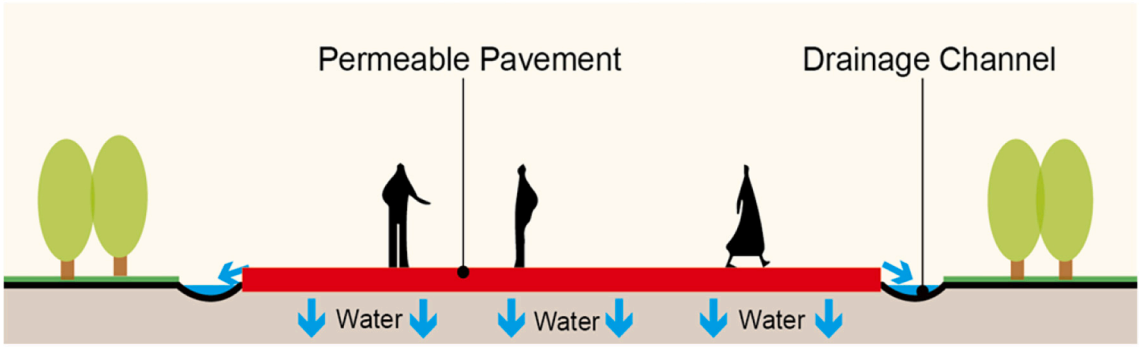
Diverse studies have highlighted a significant relationship between sound and both TSV and TCV, offering a contrast to some earlier research. Lau et al. demonstrated a notable negative correlation between TSV and sound intensity, emphasizing that TCV is also substantially affected by the outdoor sound environment (Lau and Choi, 2021). Geng et al. identified that an agreeable acoustic environment could lower subjective TSV and raise TCV (Geng et al., 2022). In certain circumstances, an increase in sound levels results in a heightened TSV and a detrimental effect on TCV (Zhou et al., 2021), as observed by Zhou et al. Furthermore, research by Guan et al. indicated that TSV values tend to rise with sound levels in neutral and warm conditions, and they documented a decline in TCV at higher sound intensities (Guan et al., 2020). These findings collectively suggest a complex interplay between auditory stimuli and thermal perception, extending the understanding of how environmental sounds influence thermal comfort and sensation.

In an intriguing departure from some established perspectives, research by scholars such as Fanger et al.(Fanger et al., 2007) has shown that variations in sound levels do not correspond with changes in TSV and TCV. Specifically, their findings suggest that noise in various environmental contexts lacks any significant correlation with shifts in TSV and TCV. Echoing this sentiment, Rahmaniar et al. also reported that ambient sound had no notable impact on TSV and TCV (Rabiyanti et al., 2017), highlighting the potential independence of thermal perception from auditory stimuli in certain environmental conditions.

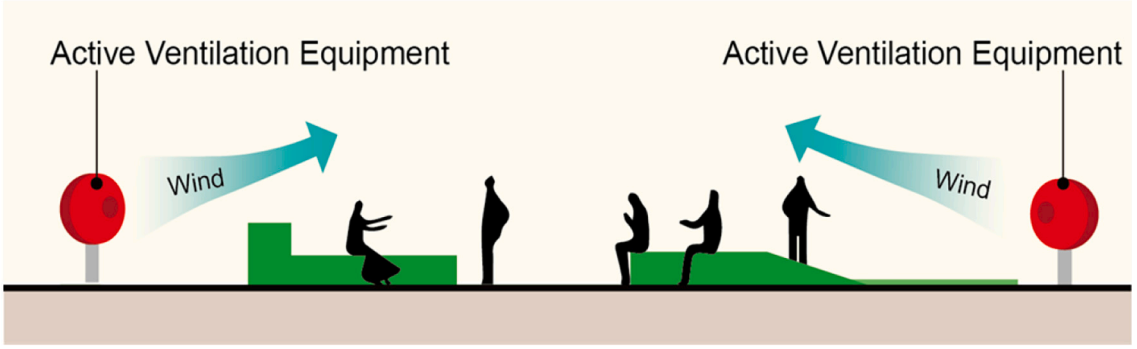
The study conducted an experimental analysis of the correlation between sound TSV and TCV in environments where sound and humidity intersect. Interestingly, the study found that changes in sound levels were linked to variations in TSV and TCV when humidity levels ranged from 25% RH to 70% RH. However, no significant impact on TSV and TCV was observed in the higher humidity range of 70% RH to 85% RH. These findings are consistent with a study conducted by Raymond et al., (2017), which found that humidity, particularly in extreme environments, primarily affects TSV and TCV. Therefore, it is essential to consider outdoor humidity levels when studying the effects of sound on TSV and TCV, especially in conditions of elevated humidity.



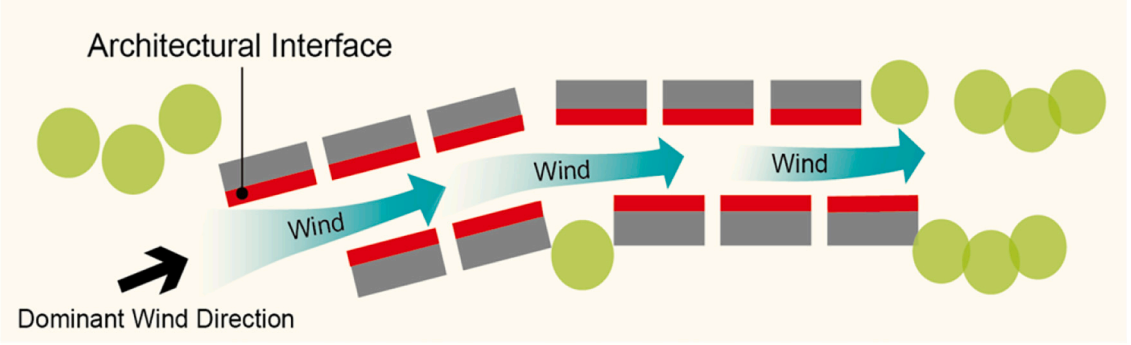
a. Noise reduction design of structures and plants



b. Design of ground permeable paving materials



c. Design of active ventilation equipment



d. Design of building interface guided ventilation

Fig. 9. Design strategies.

4.3. Design strategy

This study elucidates that the modulation of sound levels and humidity in outdoor environments exerts a significant influence on the TCV experienced by individuals. Consequently, it furnishes tailored design recommendations aimed at ameliorating outdoor acoustic conditions and mitigating atmospheric moisture levels, as delineated in Fig. 9.

Fig. 9(a) illustrates the deployment of sound insulation facilities in proximate outdoor areas without compromising the educational functionality of the campus. Wherein unsuitable for soundproofing panels, augmenting shrub density and establishing natural sound barriers prove efficacious in attenuating the outdoor acoustic milieu (Arjunan et al., 2024).

Fig. 9(b) advocates the adoption of permeable pavement over impermeable alternatives in locales characterized by elevated humidity levels (Du et al., 2023b). This strategic choice expedites surface water absorption, thereby lowering groundwater levels and consequently ameliorating environmental humidity.

Fig. 9(c) underscores the deployment of active ventilation systems near high-humidity zones across the campus precinct (Mirzaei and Haghighat, 2010). This initiative serves to accelerate outdoor airflow dynamics, facilitating expedited humidity dissipation and hence, fostering a reduction in environmental moisture content.

A pivotal facet of campus planning entails prognosticating prevailing trends. As depicted in Fig. 9(d), the strategic planning of ventilation corridors engenders a canyon effect within external building spaces. This mechanism enhances airflow dynamics and localized ventilation, thereby engendering a diminution in overall campus humidity levels (Li et al., 2024).

4.4. Limitation and future work

The current study has some limitations. Primarily, the study was restricted to university campuses in colder climates, with college students constituting the main study group.

Therefore, whether our conclusions are applicable to other climate regions and people of other age groups needs to be verified by expanding the sample range. Furthermore, the study concentrated exclusively on the influence of the sound environment and humidity on outdoor thermal comfort, excluding other factors that might also significantly affect thermal comfort, such as wind speed and solar radiation. Future investigations should strive to integrate these additional elements to provide a more encompassing perspective of outdoor thermal comfort. Additionally, the practicality and effectiveness of the design interventions proposed in this study should be further validated through quantitative numerical simulations.

5. Conclusion

We collected experimental data under different humidity ranges and different sound decibel ranges on the Xi'a University campus. This effort was specifically aimed at assessing the impact of sound and humidity environments on thermal comfort among the campus population in open spaces. The results from this data collection provide compelling evidence of the interactive effects between humidity and sound within the context of thermal comfort. The primary conclusions drawn from the study are:

(1) In the context of environments with both acoustic and humidity elements, the study revealed that a rise in humidity levels leads to an increased sensation of heat, as indicated by the upward trend of the MTSV. Within a sound decibel range of 20 to 80 dB (A) and humidity levels between 70% RH and 85% RH, the MPET exceeding 36 °C resulted in MTSV attaining and maintaining its maximum value of 3. In contrast, under the same range of sound decibels, MTSV only reached this peak value at an MPET of 43 °C, but this occurred within a lower humidity range of 25% RH to 70% RH. This demonstrates the nuanced interaction between humidity levels, sound environment, and perceived thermal sensation, particularly in varying humidity contexts.

(2) Within the context of environments incorporating both sound and humidity, the study observed a notable increase in outdoor heat perception sensitivity as sound decibel levels escalated. The analysis across the decibel ranges of 20 to 40 dB (A), 40 to 60 dB (A), and 60 to 80 dB (A) revealed that the slopes of the MTSV and MPET fitted lines were 0.1307, 0.1482, and 0.1769, respectively. Significantly, the slope in the 60–80 dB (A) range was much steeper compared to those in the 20–40 dB (A) and 40–60 dB (A) ranges. This pattern clearly illustrates that the sensitivity of MTSV to heat perception increases with higher sound decibel values, underscoring the intricate relationship between auditory stimuli and thermal sensation in outdoor environments.

(3) Within the combined influence of sound and humidity, our research highlights the imperative to consider the prevailing humidity conditions when evaluating the effects of sound decibel intervals on MTSV and MTCV. The study observed that in humidity ranges from 25% RH to 70% RH, fluctuations in sound levels significantly affected both MTSV and MTCV. However, in more elevated humidity conditions (70% RH to 85% RH), a direct correlation between sound levels and the outcomes for MTCV or MTSV was not evident. This underscores the critical role of outdoor humidity in the complex interplay between the acoustic and thermal environments, particularly in assessing the joint impact on thermal comfort and sensation.

(4) Within environments influenced by both acoustic and humidity factors, the study identified that lower sound levels may be instrumental in averting extreme thermal stress. Specifically, in lower decibel ranges, extreme thermal stress is observed at an MPET of 48, where the MTCV and MTSV reach their peak values of −3 and 3, respectively. On the other hand, in higher decibel ranges, the onset of extreme thermal stress is marked at a lower MPET of 43, with MTCV and MTSV achieving their limit values. This phenomenon, however, appears to be specific to the given conditions and necessitates further verification through

additional data collection. A key focus for future research should be to unravel the potential unique interactions between higher humidity levels in lower-decibel environments and their effects on MTSV and MTCV.

The research highlights the pivotal role played by both the acoustic environment and humidity levels in optimizing outdoor human thermal comfort. These findings are of significant value to architects, urban planners, and policymakers, offering them practical insights for the design and improvement of outdoor spaces with a focus on promoting human comfort and well-being. The intricate interplay between sound and humidity in outdoor environments necessitates meticulous consideration. Effectively addressing these factors is crucial for creating outdoor spaces that not only provide ideal thermal comfort but also positively influence the overall well-being of individuals using these spaces.

CRediT authorship contribution statement

Zeming Qin: Writing – original draft, Investigation, Data curation. **Bairu Lu:** Methodology, Conceptualization. **Wenqiang Jing:** Supervision, Investigation. **Yingdi Yin:** Validation. **Lefang Zhang:** Resources, Investigation. **Xinqi Wang:** Investigation, Data curation. **Wei Luo:** Resources, Investigation. **Jin Zhang:** Resources, Investigation. **Jing Qiu:** Investigation. **Yuting Dong:** Investigation.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

The data that has been used is confidential.

Acknowledgments

The study was supported by “Scientific Research Platform of Xi’an Eurasia University (2022XJPT01)”,“The Innovation Team of Xi’an Eurasia University (2021XJTD01)”, Xi’an Eurasia University Technical Service Special Project (OYKF-2022003), Xi’an Eurasian University School-Level Fund (2023XJZK01), Xi’an Eurasian University School-Level Fund(2023XJZK04)and Xi’an Social Science Planning Fund Project (24GL70).

References

Abbasi, A.M., Motamedzade, M., Aliabadi, M., Golmohammadi, R., Tapak, L., 2020. Combined effects of noise and air temperature on human neurophysiological responses in a simulated indoor environment. Appl. Ergon. 88 https://doi.org/10.1016/j.apergo.2020.103189.

AC, H, 1972. Thermal comfort and moderate heat stress. Br. J. Ind. Med. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1009594/.

Acero, J.A., Ruefenacht, L.A., Koh, E.J.Y., Tan, Y.S., Norford, L.K., 2022. Measuring and comparing thermal comfort in outdoor and semi-outdoor spaces in tropical Singapore. Urban Clim. 42 https://doi.org/10.1016/j.uclim.2022.101122.

Al-Atrash, F.Z., Al-Ayyoub, A., 2023. Evaluating urban outdoor thermal comfort in Jabal Al Natheef Amman. Sustainability 15 (5). https://doi.org/10.3390/su15054092.

Amaripadath, D., Rahif, R., Velickovic, M., Attia, S., 2023. A systematic review on role of humidity as an indoor thermal comfort parameter in humid climates. J. Build. Eng. 68 https://doi.org/10.1016/j.jobe.2023.106039.

Arjunan, A., Baroutaji, A., Robinson, J., Vance, A., Arafat, A., 2024. Acoustic metamaterials for sound absorption and insulation in buildings. Build. Environ. 251 https://doi.org/10.1016/j.buildenv.2024.111250.

Berglund, L.G., 1998. Comfort and humidity. ASHRAE J. 40 (8), 35. https://www.proquest.com/openview/a419ecf16879a52609bb5806cce1bf79/1?pq-origsite=gscholar&cbl=41118.

Chen, C.-P., Hwang, R.-L., Chang, S.-Y., Lu, Y.-T., 2011. Effects of temperature steps on human skin physiology and thermal sensation response. Build. Environ. 46 (11), 2387–2397. https://doi.org/10.1016/j.buildenv.2011.05.021.

Davtalab, J., Deyhimi, S.P., Dessi, V., Hafezi, M.R., Adib, M., 2020. The impact of green space structure on physiological equivalent temperature index in open space. Urban Clim. 31 https://doi.org/10.1016/j.uclim.2019.100574.

Du, M., Hong, B., Gu, C., Li, Y., Wang, Y., 2023a. Multiple effects of visual-acoustic-thermal perceptions on the overall comfort of elderly adults in residential outdoor environments. Energ. Build. 283 https://doi.org/10.1016/j.enbuild.2023.112813.

Du, X., Yang, M., Yin, Z., Fang, X., 2023b. Influence of initial abstraction ratios in NRCS-CN model on runoff estimation of permeable brick pavement affected by clogging. Water Resour. Manag. 37 (8), 3211–3225. https://doi.org/10.1007/s11269-023-03498-w.

Fanger, P.O., Breum, N.O., Jerking, E., 2007. Can colour and noise influence man’s thermal comfort? Ergonomics 20 (1), 11–18. https://doi.org/10.1080/00140137708931596.

Favero, M., Luparelli, A., Carlucci, S., 2023. Analysis of subjective thermal comfort data: a statistical point of view. Energ. Build. 281 https://doi.org/10.1016/j.enbuild.2022.112755.

Geng, Y., Hong, B., Du, M., Yuan, T., Wang, Y., 2022. Combined effects of visual-acoustic-thermal comfort in campus open spaces: a pilot study in China’s cold region. Build. Environ. 209 https://doi.org/10.1016/j.buildenv.2021.108658.

Guan, H., Hu, S., Liu, G., Zhang, L., 2020. The combined effects of temperature and noise on the comfort perceptions of young people with a normal body mass index. Sustain. Cities Soc. 54 https://doi.org/10.1016/j.scs.2019.101993.

Heaviside, C., Macintyre, H., Vardoulakis, S., 2017. The urban Heat Island: implications for health in a changing environment. Curr. Environ. Health Rep. 4 (3), 296–305. https://doi.org/10.1007/s40572-017-0150-3.

Jin, Y., Wang, F., Carpenter, M., Weller, R.B., Tabor, D., Payne, S.R., 2020. The effect of indoor thermal and humidity condition on the oldest-old people’s comfort and skin condition in winter. Build. Environ. 174 https://doi.org/10.1016/j.buildenv.2020.106790.

Juang, D.F., Lee, C.H., Yang, T., Chang, M.C., 2010. Noise pollution and its effects on medical care workers and patients in hospitals. Int. J. Environ. Sci. Technol. 7 (4), 705–716. https://doi.org/10.1007/BF03326180.
Kumar, P., Sharma, A., 2020. Study on importance, procedure, and scope of outdoor thermal comfort –a review. Sustain. Cities Soc. 61 https://doi.org/10.1016/j.scs.2020.102297.
Lai, D., Lian, Z., Liu, W., Guo, C., Liu, W., Liu, K., Chen, Q., 2020. A comprehensive review of thermal comfort studies in urban open spaces. Sci. Total Environ. 742 https://doi.org/10.1016/j.scitotenv.2020.140092.
Lam, C.K.C., Hang, J., Zhang, D., Wang, Q., Ren, M., Huang, C., 2021. Effects of short-term physiological and psychological adaptation on summer thermal comfort of outdoor exercising people in China. Build. Environ. 198 https://doi.org/10.1016/j.buildenv.2021.107877.
Lau, K.K.-L., Choi, C.Y., 2021. The influence of perceived aesthetic and acoustic quality on outdoor thermal comfort in urban environment. Build. Environ. 206 https://doi.org/10.1016/j.buildenv.2021.108333.
Li, Q., Zheng, J., Yuan, S., Zhang, L., Dong, R., Fu, H., 2024. RAV model: study on urban refined climate environment assessment and ventilation corridors construction. Build. Environ. 248 https://doi.org/10.1016/j.buildenv.2023.111080.
Ma, X., Tian, Y., Du, M., Hong, B., Lin, B., 2021. How to design comfortable open spaces for the elderly? Implications of their thermal perceptions in an urban park. Sci. Total Environ. 768 https://doi.org/10.1016/j.scitotenv.2021.144985.
Mirzaei, P.A., Haghighat, F., 2010. A novel approach to enhance outdoor air quality: pedestrian ventilation system. Build. Environ. 45 (7), 1582–1593. https://doi.org/10.1016/j.buildenv.2010.01.001.
Nicolas Pellerin, V.C., 2003. Combined effects of temperature and noise on human discomfort. Indoor Built Environ. 27 (8) https://doi.org/10.1177/1420326X17703774.
Potchter, O., Cohen, P., Lin, T.-P., Matzarakis, A., 2022. A systematic review advocating a framework and benchmarks for assessing outdoor human thermal perception. Sci. Total Environ. 833 https://doi.org/10.1016/j.scitotenv.2022.155128.
Rabiyanti, Rahmaniar, I., Putra, J.C.P., 2017. Effect of acoustic and thermal comfort to support learning process in a university. Proc. Eng. 170, 280–285. https://doi.org/10.1016/j.proeng.2017.03.027.
Raymond, C., Singh, D., Horton, R.M., 2017. Spatiotemporal patterns and synoptics of extreme wet-bulb temperature in the contiguous United States. J. Geophys. Res. Atmos. 122 (24) https://doi.org/10.1002/2017jd027140.
Sachindra, D.A., Nowosad, M., 2022. Variations in relative humidity across Poland and its possible impacts on outdoor thermal comfort: an analysis based on hourly data from 1995 to 2020. Int. J. Climatol. 42 (7), 3861–3887. https://doi.org/10.1002/joc.7449.
Yang, W., Moon, H.J., 2018. Cross-modal effects of noise and thermal conditions on indoor environmental perception and speech recognition. Appl. Acoust. 141, 1–8. https://doi.org/10.1016/j.apacoust.2018.06.019.
Yang, B., Yao, H., Yang, P., Guo, Y., Wang, F., Yang, C., Li, A., Che, L., 2022. Effects of thermal and acoustic environments on workers' psychological and physiological stress in deep underground spaces. Build. Environ. 212 https://doi.org/10.1016/j.buildenv.2022.108830.
Yao, J., Yang, F., Zhuang, Z., Shao, Y., Yuan, P.F., 2018. The effect of personal and microclimatic variables on outdoor thermal comfort: a field study in a cold season in Lujiazui CBD, Shanghai. Sustain. Cities Soc. 39, 181–188. https://doi.org/10.1016/j.scs.2018.02.025.
Yin, Y., Luo, W., Jing, W., Zhang, J., Qin, Z., Zhen, M., 2022. Combined effects of thermal-PM2.5 indicators on subjective evaluation of campus environment. Build. Environ. 222 https://doi.org/10.1016/j.buildenv.2022.109381.
Yuan, T., Hong, B., Qu, H., Liu, A., Zheng, Y., 2023. Outdoor thermal comfort in urban and rural open spaces: a comparative study in China's cold region. Urban Clim. 49 https://doi.org/10.1016/j.uclim.2023.101501.
Zhang, K., Cao, C., Chu, H., Zhao, L., Zhao, J., Lee, X., 2023a. Increased heat risk in wet climate induced by urban humid heat. Nature 617 (7962), 738–742. https://doi.org/10.1038/s41586-023-05911-1.
Zhang, S., Zhang, X., Niu, D., Fang, Z., Chang, H., Lin, Z., 2023b. Physiological equivalent temperature-based and universal thermal climate index-based adaptive-rational outdoor thermal comfort models. Build. Environ. 228 https://doi.org/10.1016/j.buildenv.2022.109900.
Zhou, X., Liu, Y., Luo, M., Zheng, S., Yang, R., Zhang, X., 2021. Overall and thermal comfort under different temperature, noise, and vibration exposures. Indoor Air 32 (1). https://doi.org/10.1111/ina.12915.
Zhu, H., Wang, H., Liu, Z., Li, D., Kou, G., Li, C., 2018. Experimental study on the human thermal comfort based on the heart rate variability (HRV) analysis under different environments. Sci. Total Environ. 616-617, 1124–1133. https://doi.org/10.1016/j.scitotenv.2017.10.208.